Block practice · Python: selection, fixing errors and short programs

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Finish an if/elif/else so each input gives the right message

Fix the errors in a short program

Write a short number program from a list of requirements

Turn a flowchart or pseudocode into Python

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Practice workbook · Finish an if/elif/else so each input gives the right message

A given program has an unfinished if statement; the student writes the conditions (relational operators, and/or/not, range boundaries) and branches so each input produces the output in a table or description.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-02

1. Question 3 · part c · 6 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

3 Programs use logic, display data and validate input.

(a) The truth table shows the initial values for A and B.

Complete the truth table to show the results of each operation.

(4)

A	B	A OR B	NOT B	(A OR B) AND (NOT B)
0	0			
0	1			
1	0			
1	1			

(b) **Figure 2** shows the intended output from a program that displays every other name in an array of star names.

```
Alasia
Castor
Electra
Gudja
Izar
Kang
Maia
Ogma
```

Figure 2

Open **Q03b** in the code editor.

Amend the code to display every other name in the array of star names.

Do not add any further functionality.

Save your code as **Q03bFINISHED** with the correct file extension for the programming language.

(3)

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6

P 7 5 7 3 7 A 0 6 1 6

1. Question 3 · part c · 6 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

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(c) A program validates a number input by the user.

Figure 3 shows the output messages based on the inputted number.

Input	Output
<empty>	You must provide a number
Any negative number	The number must be greater than zero
0	The number must be greater than zero
1 to 20	Acceptable
60 or more	Acceptable
31 to 39	Centre
30	Perfect
Any other number	No message

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Figure 3

Open **Q03c** in the code editor.

Amend the code to ensure the messages are generated correctly.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 13 marks)

1. Question 3 · part c · 6 marks · Working and answer

Working and answer

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2. Question 1 · part f · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part f only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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2. Question 1 · part f · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part f only.

(e) Describe the difference between a local variable and a global variable in a computer program.

(2)

(f) Trevor is the manager of a shoe shop.

Sales assistants can earn a bonus based on the numbers of pairs of shoes they sell and the total income for the shop each day.

Open **Q01f** in the code editor.

Amend the code to complete the 'if statement' used to produce the outputs described in the table.

Condition	Output
Shop income is more than £5000 or sales assistant has sold at least 10 pairs of shoes	Bonus is 10% of salary
Shop income is £2000 or more and sales assistant has sold at least 5 pairs of shoes	Bonus is 5% of salary

Do not add any further functionality.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(4)

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2. Question 1 · part f · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part f only.

(g) Car registration codes consist of seven characters.

The characters are two upper case letters followed by two numbers and then three upper case letters.

YH62KLV, EY09VPB and CK53PRJ are valid codes.

All codes need to be validated.

Complete the table to show **two** additional validation tests.

For each test give **one** example of erroneous data. The example that you give should fail **only** that test.

(4)

Validation test	Erroneous data
Is it 7 characters?	AB12CD

(Total for Question 1 = 16 marks)

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2. Question 1 · part f · 4 marks · Working and answer

Working and answer

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3. Question 4 · part a · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

4 Gianluca is a schoolteacher. He is in charge of a year group of pupils.

(a) He has started to create a program to analyse test results.

Open **Q04a** in the code editor.

Amend the code to complete the 'if statement' used to produce the outputs described in this table.

Condition	Output text
English test score was less than 40; Maths test score was less than 50	Student failed both tests
English test score was less than 40 or Maths test score was less than 50	Student failed one test
English test score was at least 80; Maths test score was at least 85	Student passed both tests with distinction
Other conditions	Student passed both tests

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(6)

(b) Gianluca has started to create a program to analyse pupil attendance.

He wants to use a menu system with subprograms.

Open **Q04b** in the code editor.

The program must include subprograms to:

- display the **names** of students whose percentage attendance was less than 75%
- count and display the **number** of students whose percentage attendance was 90% or higher.

Complete the program code to implement these requirements using the structure given in **Q04b**.

Do not add any further functionality.

Save your code as **Q04bFINISHED** with the correct file extension for the programming language.

(7)

(Total for Question 4 = 13 marks)

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3. Question 4 · part a · 6 marks · Working and answer

Working and answer

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4. Question 2 · part b · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

2 A book club uses computer applications.

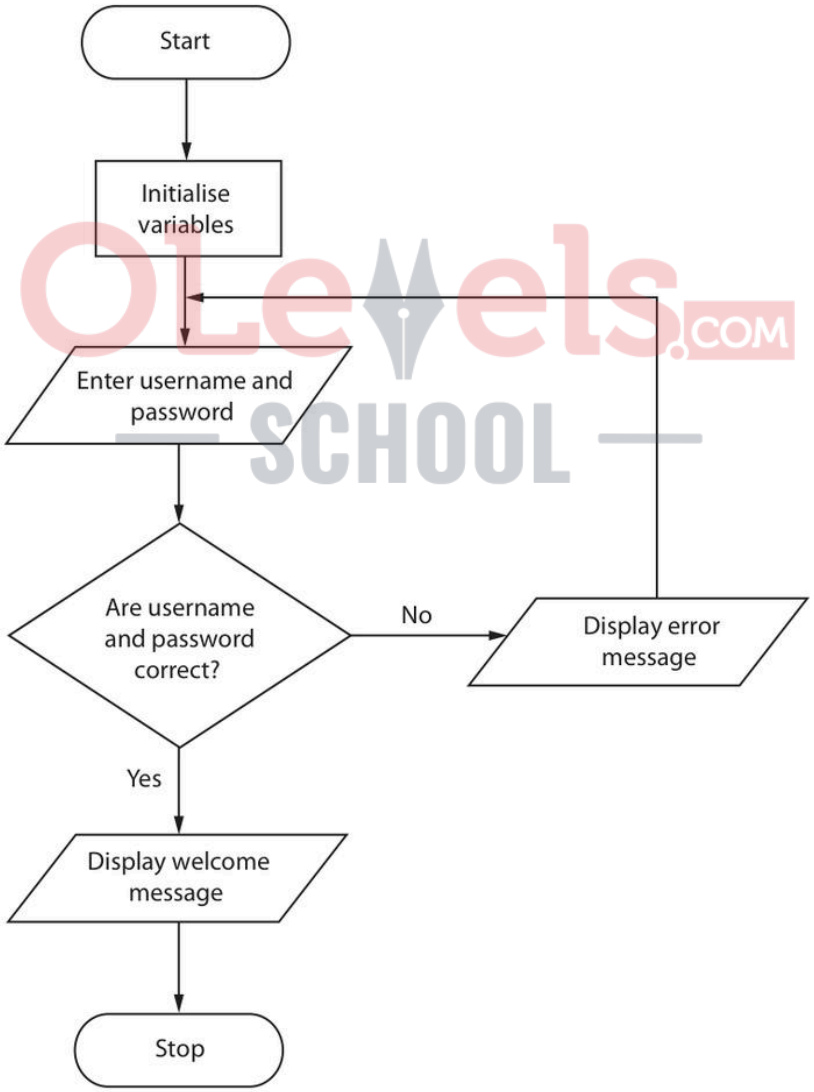
(a) The club wants a program to ensure that logins are valid.

A flowchart for a test version of the program has been designed.

The test version uses:

- a username of bard423
- a password of nX2934?

Here is the flowchart.



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4. Question 2 · part b · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

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Write a program to implement the logic in the flowchart.

Open **Q02a** in the code editor.

You must use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(11)

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4. Question 2 · part b · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

(b) The club sells books.

It needs a computer program to monitor:

- the number of books sold
- the amount of profit made.

Q02b provides a structure for the program.

Open **Q02b** in the code editor.

Amend the code to complete the If statement used to produce the outputs described in the table.

Condition	Output message
Number of books sold is at least 5 and profit made is at least 10	Sales and profit are good this week
Number of books sold is over 20 and profit made is at least 20	Sales and profit are excellent this week
Number of books sold is under 5 or profit made is under 5	Poor performance this week
All other inputs	Alert manager

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(4)

DO NOT WRITE IN THIS AREA

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6

P 6 6 4 9 0 A 0 6 1 6

4. Question 2 · part b · 4 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

(c) Boundary testing is an important part of program development.

The boundary value is the most extreme value that will pass the test.

For example, if the boundary condition is no more than 30 then the boundary test data would be 30.

Complete the test table to show the boundary test data that could be used for each test.

(3)

	Test data	Expected output
booksSold		Poor performances this week
profit		
booksSold		Sales and profit are good this week
profit		
booksSold		Sales and profit are excellent this week
profit		

(Total for Question 2 = 18 marks)

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4. Question 2 · part b · 4 marks · Working and answer

Working and answer

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5. Question 1 · part g · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part g only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

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DO NOT WRITE IN THIS AREA

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5. Question 1 · part g · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part g only.

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(d) Describe **one** difference between the data used in boundary testing and the data used in erroneous testing.

(2)

(e) A program should output the value 2

However, there is an error in the code and the actual output is 4

Name this type of error.

(1)

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DO NOT WRITE IN THIS AREA

(f) Open **Q01f** in the code editor.

The program should allow the input of the length of the side of a square and output the area of the square.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(3)

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(g) A program is needed that will accept the input of a letter and compare it with a stored letter.

It will check whether the letter comes earlier in the alphabet, later in the alphabet or is the same letter as the stored letter.

It will output the letter and the stored letter with an appropriate message.

Open **Q01g** in the code editor.

Amend the code to complete the if statement used to produce the output.

You must use the structure given in **Q01g** to complete the program.

Do not add any further functionality.

Save your code as **Q01gFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 1 = 14 marks)

5. Question 1 · part g · 4 marks · Working and answer

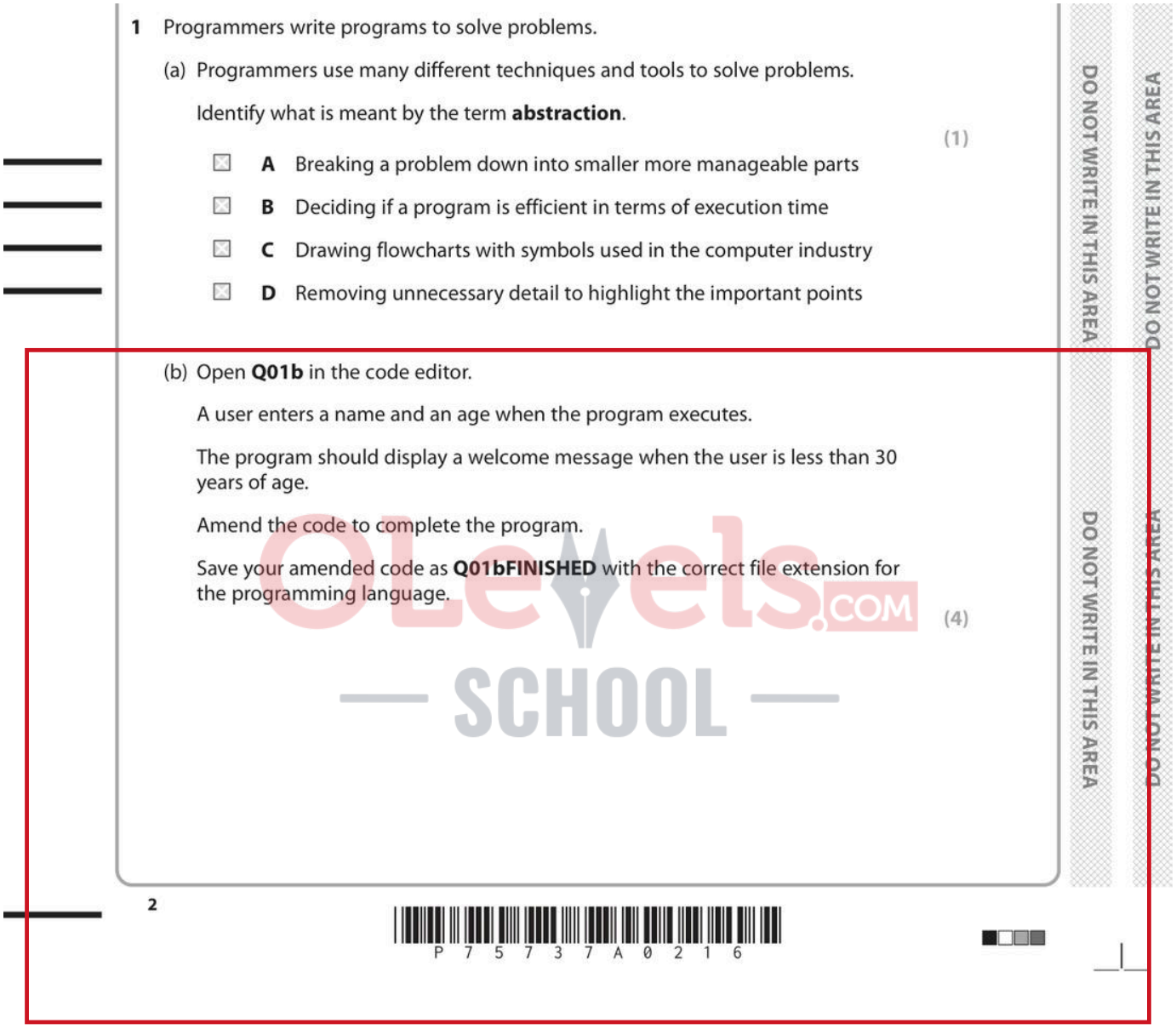
Working and answer

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6. Question 1 · part b · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.



1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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2

P 7 5 7 3 7 A 0 2 1 6

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6. Question 1 · part b · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

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6. Question 1 · part b · 4 marks · Working and answer

Working and answer

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Practice workbook · Fix the errors in a short program

A short supplied Python program that should do a stated job contains a stated number (usually three, sometimes four) of errors; the student corrects them in the editor and saves the FINISHED file.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-01

1. Question 2 · part c · 3 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

2 Programmers design, correct and test programs.

(a) Programmers design programs.

(i) Identify the term for a step-by-step description to complete a task.

(1)

- ☒ **A** Algorithm
- ☒ **B** Computational thinking
- ☒ **C** Decomposition
- ☒ **D** Pseudocode

(ii) A constant is a memory location whose value does not change during program execution.

Give the name for a memory location whose value can change during program execution.

(1)

(b) Programs can have runtime errors.

(i) State what is meant by the term **runtime error**.

(1)

(ii) Give **one** example of a runtime error.

(1)

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1. Question 2 · part c · 3 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part c only.

(c) Open **Q02c** in the code editor.

The program generates a random number for the width of a rectangle.

The length of the rectangle is always 4

The program should calculate the perimeter of the rectangle.

The program should display the width, the length and the perimeter of the rectangle.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q02cFINISHED** with the correct file extension for the programming language.

(3)

(d) **Figure 1** shows part of an online form displayed in a browser.

Memorable word		(2-8 letters)
Date	31-01-2019	

Figure 1

Complete the table to show **two** examples of erroneous, **one** example of normal and **one** example of boundary test data for the **memorable word** field.

Each example must be different.

(4)

Erroneous	
Erroneous	
Normal	
Boundary	

(Total for Question 2 = 11 marks)

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1. Question 2 · part c · 3 marks · Working and answer

Working and answer

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2. Question 1 · part c · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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2. Question 1 · part c · 3 marks · Working and answer

Working and answer

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3. Question 1 · part c · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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3. Question 1 · part c · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

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(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

3. Question 1 · part c · 3 marks · Working and answer

Working and answer

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4. Question 1 · part c · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

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2

P 6 6 4 9 0 A 0 2 1 6

4. Question 1 · part c · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

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(d) Raoul has written a program that counts the number of vowels in a sentence.

Open **Q01d** in the code editor.

Amend the code by adding a suitable comment (you may need to add more than one comment to a line):

(i) at the end of the line where there is a **relational operator**

(1)

(ii) at the end of a line where **iteration** starts

(1)

(iii) at the end of the line where **selection** starts

(1)

(iv) at the end of a line where a **data structure** is initialised.

(1)

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(Total for Question 1 = 10 marks)

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Working and answer

5. Question 1 · part d · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

1 Programmers write code to solve problems.

(a) Give **one** technique that programmers use to make their code easy to read and understand.

(1)

.....

(b) Programs may contain logic, syntax or runtime errors.

A program has three errors.

These may be logic, syntax or runtime errors.

Complete the table by adding a tick (✓) for **each** error description to show whether the error is a logic, syntax or runtime error.

(3)

Error description	Logic	Syntax	Runtime
Divide by 0			
Use x instead of * to multiply			
Subtract 10 from 2 instead of 2 from 10			

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5. Question 1 · part d · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

(c) **Figure 1** shows a programming construct in a flowchart.

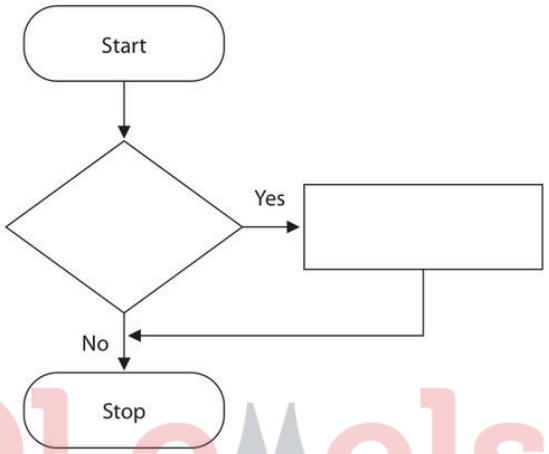


Figure 1

Identify the name of the programming construct.

(1)

- ☒ **A** Array
- ☒ **B** Iteration
- ☒ **C** Operator
- ☒ **D** Selection

(d) Open **Q01d** in the code editor.

The program inputs two numbers, compares them and outputs a suitable message.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(3)

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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5. Question 1 · part d · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

(e) A program is needed to calculate and print a multiplication table for a number that is input by the user.

Figure 2 shows the display when the number 2 is input.

```
Enter the number: 2
1 x 2 is 2
2 x 2 is 4
3 x 2 is 6
4 x 2 is 8
5 x 2 is 10
6 x 2 is 12
7 x 2 is 14
8 x 2 is 16
9 x 2 is 18
10 x 2 is 20
11 x 2 is 22
12 x 2 is 24
```

Figure 2

Open **Q01e** in the code editor.

Write the program.

You **must** use the structure given in the file **Q01e** to complete the program.

Do not add any further functionality.

Save your code as **Q01eFINISHED** with the correct file extension for the programming language.

(5)

(Total for Question 1 = 13 marks)

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5. Question 1 · part d · 3 marks · Working and answer

Working and answer

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6. Question 1 · part f · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part f only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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6. Question 1 · part f · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part f only.

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(d) Describe **one** difference between the data used in boundary testing and the data used in erroneous testing.

(2)

(e) A program should output the value 2

However, there is an error in the code and the actual output is 4

Name this type of error.

(1)

DO NOT WRITE IN THIS AREA

(f) Open **Q01f** in the code editor.

The program should allow the input of the length of the side of a square and output the area of the square.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(3)

DO NOT WRITE IN THIS AREA

(g) A program is needed that will accept the input of a letter and compare it with a stored letter.

It will check whether the letter comes earlier in the alphabet, later in the alphabet or is the same letter as the stored letter.

It will output the letter and the stored letter with an appropriate message.

Open **Q01g** in the code editor.

Amend the code to complete the if statement used to produce the output.

You must use the structure given in **Q01g** to complete the program.

Do not add any further functionality.

Save your code as **Q01gFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 1 = 14 marks)

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6. Question 1 · part f · 3 marks · Working and answer

Working and answer

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7. Question 2 · part b.ii · 4 marks

2023 JUN · Paper 02

Further practice · Attempt part b.ii only.

2 Solutions to problems are made up of many different components.

(a) Programmers use subprograms when developing code.

(i) Give **two** reasons to use **subprogram libraries** when developing code.

(2)

1 .....

.....

2 .....

.....

(ii) Identify which **one** of these must return a result.

(1)

- ☒ **A** Function
- ☒ **B** Iteration
- ☒ **C** Procedure
- ☒ **D** Selection

(iii) State what is meant by the term **local variable**.

(1)

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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7. Question 2 · part b.ii · 4 marks

2023 JUN · Paper 02

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) A school is planning a fish and chip dinner for the students and their families.

(i) Tickets are only sold to families.

A family is one to four adults and one to six children.

A program is being written to help manage ticket sales.

Complete the table to show examples of normal, boundary and erroneous numeric test data for the program.

(3)

	Adult	Children
Normal		
Boundary		
Erroneous		

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(ii) The kitchen staff use a program to determine whether they have enough chips or how many more they need to order.

Open **Q02bii** in the code editor.

There are **four** errors in the code.

Amend the code to correct the errors.

Use this test data to help you find the errors.

Chips in stock (kilograms)	Number of adults	Number of children	Expected output
19	100	150	Stocks available
15	100	150	Order 4.0 kilograms

Save your amended code as **Q02biiFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 2 = 11 marks)

7. Question 2 · part b.ii · 4 marks · Working and answer

Working and answer

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8. Question 1 · part b.ii · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part b.ii only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8. Question 1 · part b.ii · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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8. Question 1 · part b.ii · 3 marks · Working and answer

Working and answer

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Practice workbook · Write a short number program from a list of requirements

From a bullet list of requirements (no design given) the student writes a short numeric program: inputs converted to numbers, calculations with arithmetic/modulus operators, selection for bands or tests, and count- or condition-controlled loops, including stepping through an array.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-05

1. Question 3 · part whole · 6 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.

3 Manjit sells copies of a science textbook to schools.

She needs a program to process textbook orders. It must:

- accept the input of the number of textbooks required
- generate the price per textbook
- generate the total cost of the order
- display the price per textbook and the total cost of the order.

The price of one textbook depends on the number ordered:

Quantity	Price per textbook
1–5	£20.00
6–9	£15.00
10 or more	£12.00

Open **Q03** in the code editor.

Write the program.

You must use the structure given in **Q03** to complete the program.

Do not add any further functionality.

Save your code as **Q03FINISHED** with the correct file extension for the programming language.

(Total for Question 3 = 6 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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1. Question 3 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 3 · part c · 6 marks

2019 JUN · Paper 02

Further practice · Attempt part c only.

3 Different algorithms can be used to manipulate data.

(a) **Email.txt** contains a list of email addresses.

Open **Q03a** in the code editor.

Write a program to implement these requirements.

The code must:

- check each email address to ensure it contains the '@' symbol.
- write email addresses that do not contain the '@' symbol to an **Error.txt** file.

You must use the structure given in the file **Q03a** to complete the program.

Do not add further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(6)

(b) Explain **one** drawback of using the merge sort algorithm to sort large data sets.

(2)

(c) Open **Q03c** in the code editor.

Write a program to display the square and cube of a number between 1 and 50 entered by a user.

The code must:

- ask the user to enter a number between 1 and 50 inclusive
- display the number, the square of the number and the cube of the number, with appropriate labels
- stop when a number outside the range 1 to 50 is entered.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 14 marks)

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2. Question 3 · part c · 6 marks · Working and answer

Working and answer

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3. Question 2 · part b · 7 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

2 Rama wants some computer programs for his daughter, Ayomi.

(a) Rama wants a program to calculate the area of a triangle.

(area of a triangle = ½ base x height)

This pseudocode contains the logic required to create the program.

```
1 #  Initialise variables
2
3 SET base TO 50
4 SET heightChk TO TRUE
5
6 #  Print prompt and take input from user
7
8 WHILE heightChk DO
9     SEND "Enter the height (between 1 and 100): " TO DISPLAY
10    RECEIVE height FROM (INTEGER) KEYBOARD
11    IF (height >= 1 AND height <= 100)
12    THEN heightChk = FALSE
13    END IF
14 END WHILE
15
16 #  Calculate and print out values
17
18 SET area TO 0.5 * base * height
19
20 SEND ("Base of triangle is: " , base) TO DISPLAY
21 SEND ("Height of triangle is: " , height) TO DISPLAY
22 SEND ("Area of triangle is: " , area) TO DISPLAY
```

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

3. Question 2 · part b · 7 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

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Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

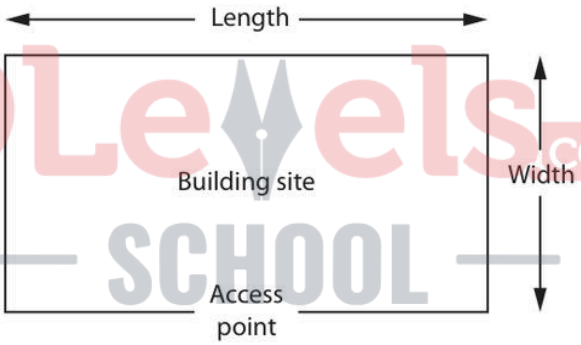
You must use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

(b) Ayomi wants a program that will calculate the number of fence panels required to surround building sites such as the one shown in the diagram (not to scale).



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Each building site is a rectangle.

Each fence panel is 1 metre long.

A gap of 4 metres is needed to access each site. The location of the gap is not significant.

The user will input the length and width of the building site as whole numbers of metres.

The program will output the minimum number of full panels needed.

Q02b provides a structure for the program.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to write the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(7)

(Total for Question 2 = 17 marks)

3. Question 2 · part b · 7 marks · Working and answer

Working and answer

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4. Question 1 · part e · 5 marks

2021 NOV · Paper 02

Further practice · Attempt part e only.

1 Programmers write code to solve problems.

(a) Give **one** technique that programmers use to make their code easy to read and understand.

(1)

.....

(b) Programs may contain logic, syntax or runtime errors.

A program has three errors.

These may be logic, syntax or runtime errors.

Complete the table by adding a tick (✓) for **each** error description to show whether the error is a logic, syntax or runtime error.

(3)

Error description	Logic	Syntax	Runtime
Divide by 0			
Use x instead of * to multiply			
Subtract 10 from 2 instead of 2 from 10			

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4. Question 1 · part e · 5 marks

2021 NOV · Paper 02

Further practice · Attempt part e only.

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(c) **Figure 1** shows a programming construct in a flowchart.

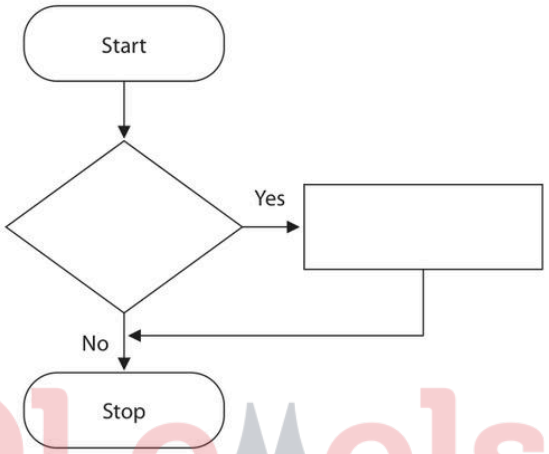


Figure 1

Identify the name of the programming construct.

(1)

- ☒ **A** Array
- ☒ **B** Iteration
- ☒ **C** Operator
- ☒ **D** Selection

(d) Open **Q01d** in the code editor.

The program inputs two numbers, compares them and outputs a suitable message.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(3)

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4. Question 1 · part e · 5 marks

2021 NOV · Paper 02

Further practice · Attempt part e only.

(e) A program is needed to calculate and print a multiplication table for a number that is input by the user.

Figure 2 shows the display when the number 2 is input.

```
Enter the number: 2
1 x 2 is 2
2 x 2 is 4
3 x 2 is 6
4 x 2 is 8
5 x 2 is 10
6 x 2 is 12
7 x 2 is 14
8 x 2 is 16
9 x 2 is 18
10 x 2 is 20
11 x 2 is 22
12 x 2 is 24
```

Figure 2

Open **Q01e** in the code editor.

Write the program.

You **must** use the structure given in the file **Q01e** to complete the program.

Do not add any further functionality.

Save your code as **Q01eFINISHED** with the correct file extension for the programming language.

(5)

(Total for Question 1 = 13 marks)

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4. Question 1 · part e · 5 marks · Working and answer

Working and answer

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5. Question 3 · part c · 6 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

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5. Question 3 · part c · 6 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

(b) Explain **one** reason why parameters are used to pass data into subprograms.

(2)

(c) A leap year is a year that has 366 days rather than 365 days. A leap year usually occurs every four years.

A year that is a leap year:

- can be exactly divided by 400

Or

- can be exactly divided by 4, but **not** exactly divided by 100.

These years are leap years: 2000, 2012 and 2020.

These years are not leap years: 1700, 1900 and 2021.

A program is needed that will accept the input of a year and output whether it is a leap year or not.

Open **Q03c** in the code editor.

Write the program.

You **must** use the structure given in **Q03c**.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 15 marks)

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5. Question 3 · part c · 6 marks · Working and answer

Working and answer

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6. Question 3 · part b · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

3 Programs use logic, display data and validate input.

(a) The truth table shows the initial values for A and B.

Complete the truth table to show the results of each operation.

(4)

A	B	A OR B	NOT B	(A OR B) AND (NOT B)
0	0			
0	1			
1	0			
1	1			

(b) **Figure 2** shows the intended output from a program that displays every other name in an array of star names.

```
Alasia
Castor
Electra
Gudja
Izar
Kang
Maia
Ogma
```

Figure 2

Open **Q03b** in the code editor.

Amend the code to display every other name in the array of star names.

Do not add any further functionality.

Save your code as **Q03bFINISHED** with the correct file extension for the programming language.

(3)

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6. Question 3 · part b · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

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(c) A program validates a number input by the user.

Figure 3 shows the output messages based on the inputted number.

Input	Output
<empty>	You must provide a number
Any negative number	The number must be greater than zero
0	The number must be greater than zero
1 to 20	Acceptable
60 or more	Acceptable
31 to 39	Centre
30	Perfect
Any other number	No message

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Figure 3

Open **Q03c** in the code editor.

Amend the code to ensure the messages are generated correctly.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 13 marks)

6. Question 3 · part b · 3 marks · Working and answer

Working and answer

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Practice workbook · Turn a flowchart or pseudocode into Python

The algorithm is already designed - as numbered pseudocode, a flowchart or a trace table plus requirements - and the student implements its logic in Python inside a given structure (5-11 marks).

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-CODE-04

1. Question 3 · part c · 10 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

3 Algorithms can be represented in flowcharts, pseudocode or program code.

(a) Trace tables can be used with flowcharts or pseudocode.

Give **two** characteristics of a trace table.

(2)

1 .....

.....

2 .....

.....

(b) An algorithm has been written to validate numbers entered by the user.

Figure 2 shows the pseudocode for the algorithm.

```
1  RECEIVE num1 FROM (INTEGER) KEYBOARD
2  RECEIVE num2 FROM (INTEGER) KEYBOARD
3  IF (((num1 < 16) AND (num2 < 23)) OR (num2 = 13)) THEN
4      SEND "State 1" TO DISPLAY
5  ELSE
6      IF ((num2 > 12) AND (num1 > 20)) THEN
7          SEND "State 2" TO DISPLAY
8      ELSE
9          IF ((num1 = 88) OR NOT (num2 = 18)) THEN
10             SEND "State 3" TO DISPLAY
11         ELSE
12             SEND "State 4" TO DISPLAY
13         END IF
14     END IF
15 END IF
```

Figure 2

Complete the table to show the output for each set of inputs.

(3)

num1	num2	Output
88	18	
17	18	
12	19	

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1. Question 3 · part c · 10 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

(c) A program is required to calculate the result of raising one integer (the base) to the power of another (the exponent).

Figure 3 shows a flowchart for the algorithm.

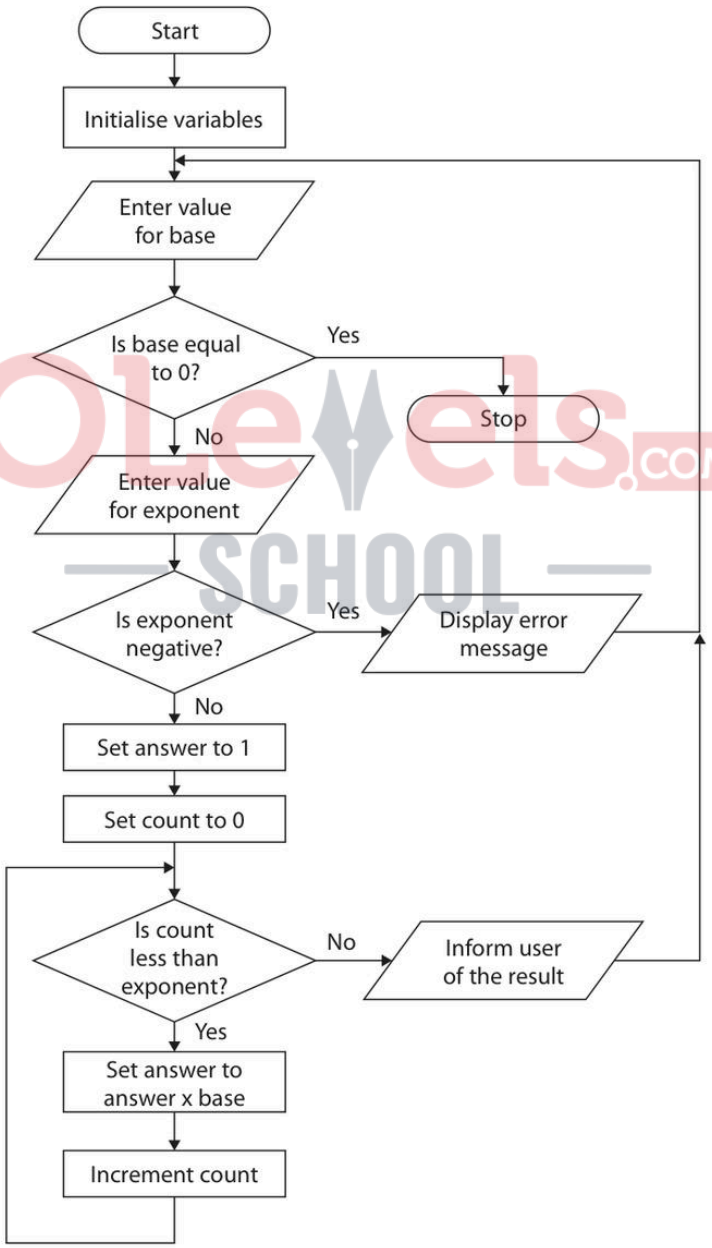


Figure 3

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1. Question 3 · part c · 10 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part c only.

The program has these requirements:

- outputs meaningful error messages
- outputs the final answer with the base and the exponent.

Open **Q03c** in the code editor.

Write a program to implement the logic in the flowchart.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(10)

(Total for Question 3 = 15 marks)

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1. Question 3 · part c · 10 marks · Working and answer

Working and answer

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2. Question 2 · part a · 10 marks

2019 JUN · Paper 02

Further practice · Attempt part a only.

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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2. Question 2 · part a · 10 marks

2019 JUN · Paper 02

Further practice · Attempt part a only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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6

P 6 1 8 8 1 R A 0 6 1 6

2. Question 2 · part a · 10 marks · Working and answer

Working and answer

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3. Question 2 · part a · 10 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

2 Rama wants some computer programs for his daughter, Ayomi.

(a) Rama wants a program to calculate the area of a triangle.

(area of a triangle = ½ base x height)

This pseudocode contains the logic required to create the program.

```
1 #  Initialise variables
2
3 SET base TO 50
4 SET heightChk TO TRUE
5
6 #  Print prompt and take input from user
7
8 WHILE heightChk DO
9     SEND "Enter the height (between 1 and 100): " TO DISPLAY
10    RECEIVE height FROM (INTEGER) KEYBOARD
11    IF (height >= 1 AND height <= 100)
12    THEN heightChk = FALSE
13    END IF
14 END WHILE
15
16 #  Calculate and print out values
17
18 SET area TO 0.5 * base * height
19
20 SEND ("Base of triangle is: " , base) TO DISPLAY
21 SEND ("Height of triangle is: " , height) TO DISPLAY
22 SEND ("Area of triangle is: " , area) TO DISPLAY
```

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3. Question 2 · part a · 10 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

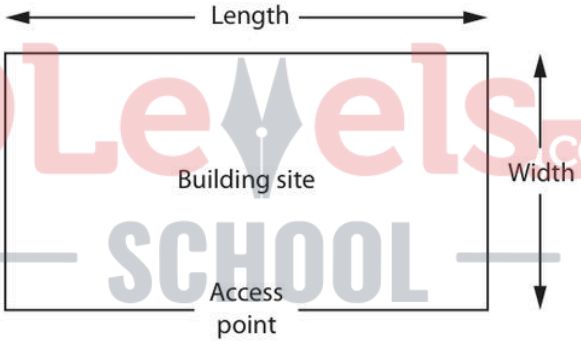
You must use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

(b) Ayomi wants a program that will calculate the number of fence panels required to surround building sites such as the one shown in the diagram (not to scale).



Each building site is a rectangle.

Each fence panel is 1 metre long.

A gap of 4 metres is needed to access each site. The location of the gap is not significant.

The user will input the length and width of the building site as whole numbers of metres.

The program will output the minimum number of full panels needed.

Q02b provides a structure for the program.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to write the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(7)

(Total for Question 2 = 17 marks)

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3. Question 2 · part a · 10 marks · Working and answer

Working and answer

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4. Question 2 · part a · 11 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

2 A book club uses computer applications.

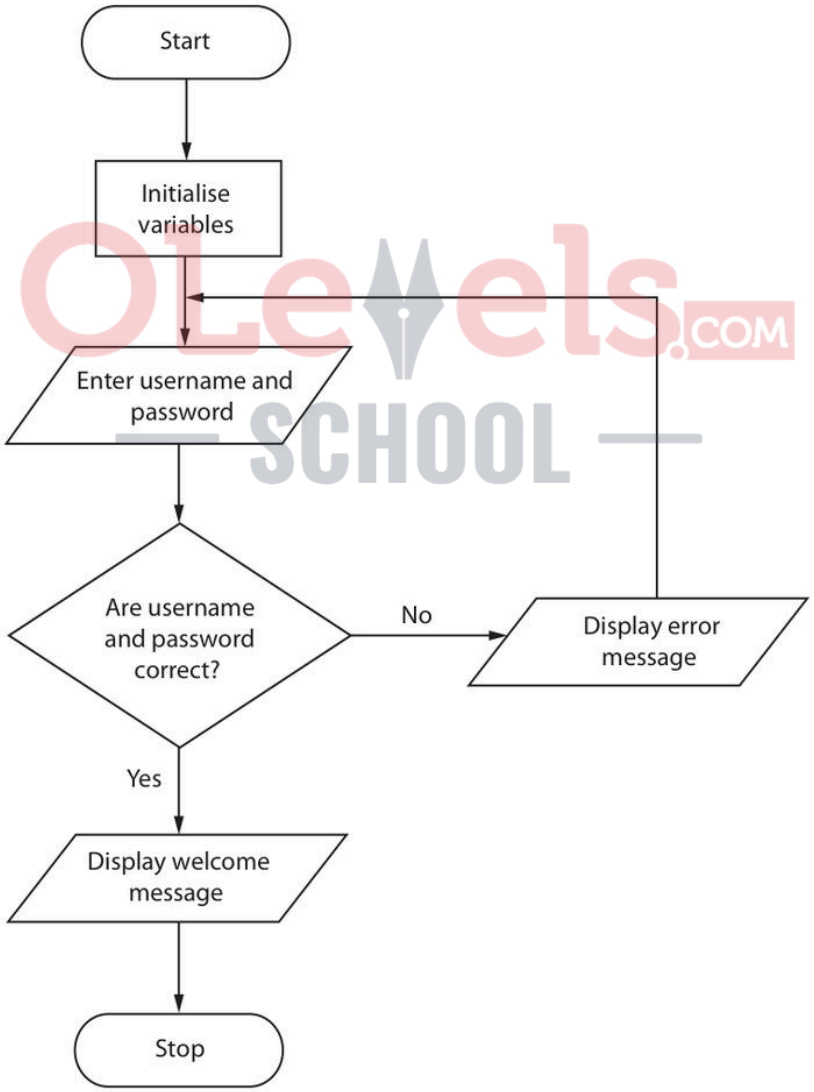
(a) The club wants a program to ensure that logins are valid.

A flowchart for a test version of the program has been designed.

The test version uses:

- a username of bard423
- a password of nX2934?

Here is the flowchart.



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4. Question 2 · part a · 11 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

Write a program to implement the logic in the flowchart.

Open **Q02a** in the code editor.

You must use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(11)

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4. Question 2 · part a · 11 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

(b) The club sells books.

It needs a computer program to monitor:

- the number of books sold
- the amount of profit made.

Q02b provides a structure for the program.

Open **Q02b** in the code editor.

Amend the code to complete the If statement used to produce the outputs described in the table.

Condition	Output message
Number of books sold is at least 5 and profit made is at least 10	Sales and profit are good this week
Number of books sold is over 20 and profit made is at least 20	Sales and profit are excellent this week
Number of books sold is under 5 or profit made is under 5	Poor performance this week
All other inputs	Alert manager

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(4)

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P 6 6 4 9 0 A 0 6 1 6

4. Question 2 · part a · 11 marks · Working and answer

Working and answer

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5. Question 2 · part a · 10 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

2 Estelle is learning about sorting and searching algorithms.

(a) Estelle's friend has given her a partially completed program to carry out a binary search.

This pseudocode contains the logic required to complete the program.

```
11 RECEIVE item FROM (INTEGER) KEYBOARD
12 WHILE start <= end AND NOT found DO
13      SET middle TO (start + end) DIV 2
14      IF numberList[middle] = item THEN
15       SET found TO TRUE
16      ELSE
17       IF item < numberList[middle] THEN
18        end = middle - 1
19       ELSE
20        start = middle + 1
21       END IF
22      END IF
23      SET count TO count + 1
24 END WHILE
```

Open file **Q02a** in the code editor.

Amend the code to complete the program.

You **must** use the structure given in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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5. Question 2 · part a · 10 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

(b) **Figure 3** shows an array of names.

0	1	2	3	4	5	6	7
Yekatrina	Juan	George	Elija	Christine	Sangeetha	Melanie	Shu-fen

Figure 3

Explain why a binary search algorithm would not find 'Juan' in the array.

(4)

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5. Question 2 · part a · 10 marks · Working and answer

Working and answer

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6. Question 2 · part b · 11 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

A prime number is a whole number, larger than one, that can only be divided by one and itself with no remainder.

This pseudocode contains the logic required to complete the program.

```
FUNCTION checkPrime(pNumber)
BEGIN FUNCTION
    IF pNumber = 1 THEN
        check = False
    ELSE
        check = True
        FOR count FROM 2 TO pNumber DO
            IF pNumber MOD count = 0 THEN
                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

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6. Question 2 · part b · 11 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

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(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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6. Question 2 · part b · 11 marks · Working and answer

Working and answer

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7. Question 3 · part c · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

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3 A youth club organises age-specific events for its members.

A program is used to find members of the correct age.

(a) The program first sorts the list of members by age.

(i) A bubble sort algorithm is used to sort the list into ascending order.

The table shows a list of ages that must be sorted.

Complete the table to show how the algorithm sorts the list.

You may not need to use all the rows.

(2)

9	11	12	19	14	18	15	17

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(ii) State the number of swaps made in the first pass.

(1)

.....

(iii) State the **least** number of passes needed to complete the sort algorithm for this list.

(1)

.....

7. Question 3 · part c · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

(b) The youth club stores a list of sports equipment.

The list contains over 1000 items, sorted in ascending order.

State and justify the **most** appropriate search algorithm to find an item in the list.

(2)

Search algorithm

.....

Justification

.....

.....

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7. Question 3 · part c · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

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(c) A program is required to calculate and display the average age of members taking part in an activity.

Figure 3 shows a flowchart for the algorithm.

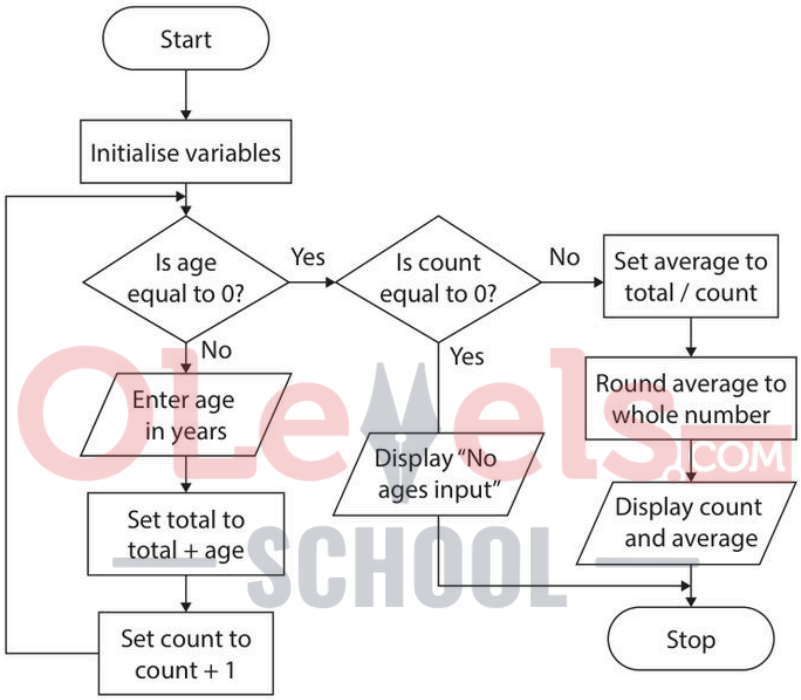


Figure 3

Open **Q03c** in the code editor.

Write a program to implement the logic of the flowchart.

Use the data given in **Figure 4** to test your program.

Test data	Expected output
7 8 9 10 11 0	You entered 5 ages; the average age is 9.
16 16 14 0	You entered 3 ages; the average age is 15.
0	No ages input

Figure 4

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(5)

(Total for Question 3 = 11 marks)

7. Question 3 · part c · 5 marks · Working and answer

Working and answer

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8. Question 4 · part c · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

4 Eli is designing a three-throw dice game to play on a computer.

He uses abstraction and decomposition to help him.

(a) **Figure 5** shows how Eli uses comments to plan code before writing it.

Eli's preferred language uses a double-slash // to write comments.

```
1  // roll dice 3 times
2
3     // display number rolled
4
5     // update subtotal
6
7     // check for roll of 3 or 6 and increment booster
8
9  // calculate and output final score
10
```

Figure 5

State why this is a type of decomposition.

(1)

(b) Eli decides to use a library subprogram to generate random numbers that model a dice throw.

Explain why this is an example of abstraction.

(2)

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P 7 8 6 3 9 A 0 1 0 2 0

8. Question 4 · part c · 8 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

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(c) Eli's program has these requirements.

- A roll of the dice is a random number between 1 and 6.
- The dice is rolled three times.
- The numbers rolled are added to a subtotal.
- The final score is calculated by multiplying the subtotal by a booster value.
- The booster value starts at 1.
- Each time a 3 or 6 is rolled, the booster value is incremented by 1.
- The game outputs the value of each roll, and the final score.

Figure 6 shows a trace table for one execution of the program.

Output	Subtotal	Booster
Rolled: 3	0 + 3 = 3	3 rolled, so incremented to 2
Rolled: 2	3 + 2 = 5	Neither 3 nor 6 rolled
Rolled: 6	5 + 6 = 11	6 rolled, so incremented to 3
Score 33	Subtotal (11) * Booster (3) = 33	

Figure 6

Open **Q04c** in the code editor.

Write the code to create this program.

You must use the layout, variables and constants given in **Q04c**.

Do not add any further functionality.

Save your code as **Q04cFINISHED** with the correct file extension for the programming language.

(8)

(Total for Question 4 = 11 marks)

8. Question 4 · part c · 8 marks · Working and answer

Working and answer

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